

Integration & Applications

ANSWER KEY



Basic antiderivatives (the power rule)

- 1 Find $\int x^4 dx$.
 $\frac{x^5}{5} + C.$
 - 2 Find $\int 3x^2 dx$.
 $x^3 + C.$
 - 3 Find $\int \sqrt{x} dx$.
 $\int x^{1/2} dx = \frac{2}{3}x^{3/2} + C.$
 - 4 Find $\int \frac{1}{x^3} dx$.
 $\int x^{-3} dx = -\frac{1}{2}x^{-2} + C = -\frac{1}{2x^2} + C.$
-

Antiderivatives of polynomials

- 5 Find $\int (4x^3 - 6x + 5) dx$.
 $x^4 - 3x^2 + 5x + C.$
 - 6 Find $\int (2x + 7) dx$.
 $x^2 + 7x + C.$
 - 7 Find $\int (x^2 - 4x + 3) dx$.
 $\frac{x^3}{3} - 2x^2 + 3x + C.$
-

Antiderivatives of trigonometric and exponential functions

- 8 Find $\int \cos x dx$.
 $\sin x + C.$
 - 9 Find $\int \sin x dx$.
 $-\cos x + C.$
 - 10 Find $\int e^x dx$.
 $e^x + C.$
-

Evaluating definite integrals

11 Evaluate $\int_0^2 3x^2 dx$.
 $[x^3]_0^2 = 8$.

12 Evaluate $\int_1^3 2x dx$.
 $[x^2]_1^3 = 9 - 1 = 8$.

13 Evaluate $\int_0^4 x dx$.
 $\left[\frac{x^2}{2}\right]_0^4 = 8$.

14 Evaluate $\int_{-1}^2 3x^2 dx$.
 $[x^3]_{-1}^2 = 8 - (-1) = 9$.

Definite integrals with polynomial integrands

15 Evaluate $\int_0^2 (x^3 + x) dx$.
 $\left[\frac{x^3}{3} + \frac{x^2}{2}\right]_0^2 = \frac{8}{3} + 2 = \frac{14}{3}$.

16 Evaluate $\int_1^3 (2x^2 - 3x + 1) dx$.
 $F(x) = \frac{2x^3}{3} - \frac{3x^2}{2} + x$. $F(3) = 18 - 13.5 + 3 = 7.5$. $F(1) = 0.667 - 1.5 + 1 = 0.167$. Integral
 $= 7.5 - 0.167 = \frac{14}{3}$.

17 Evaluate $\int_0^1 (6x^2 - 4x + 2) dx$.
 $[2x^3 - 2x^2 + 2x]_0^1 = (2 - 2 + 2) - 0 = 2$.

18 Evaluate $\int_{-2}^1 (x^2 - x) dx$.
 $F(x) = \frac{x^3}{3} - \frac{x^2}{2}$. $F(1) = -\frac{1}{6}$. $F(-2) = -\frac{8}{3} - 2 = -\frac{14}{3}$. Integral $= -\frac{1}{6} + \frac{14}{3} = \frac{9}{2}$.

Definite integrals of trigonometric and exponential functions

19 Evaluate $\int_0^{\pi/2} \cos x dx$.
 $[\sin x]_0^{\pi/2} = 1 - 0 = 1$.

20 Evaluate $\int_0^{\pi} \sin x dx$.
 $[-\cos x]_0^{\pi} = -\cos(\pi) - (-\cos(0)) = 1 - (-1) = 2$.

21 Evaluate $\int_0^1 e^x dx$.
 $[e^x]_0^1 = e - 1 \approx 1.72$.

Area under a curve

- 22 Find the area under $y = x^2$ from $x = 0$ to $x = 3$, using a definite integral.

$$\text{Area} = \int_0^3 x^2 dx = \left[\frac{x^3}{3} \right]_0^3 = \frac{27}{3} = 9.$$
- 23 Find the area under $y = 4 - x^2$ from $x = -2$ to $x = 2$.

$$F(x) = 4x - \frac{x^3}{3}. F(2) = 8 - \frac{8}{3} = \frac{16}{3}. F(-2) = -8 + \frac{8}{3} = -\frac{16}{3}. \text{Area} = \frac{16}{3} - \left(-\frac{16}{3}\right) = \frac{32}{3}.$$
- 24 Find the area under $y = 3\sqrt{x}$ from $x = 0$ to $x = 4$.

$$\int 3x^{1/2} dx = 2x^{3/2}. \text{At } x = 4: 2(8) = 16. \text{At } x = 0: 0. \text{Area} = 16.$$
-

Signed area and regions below the axis

- 25 Evaluate $\int_0^2 (x - 2) dx$ and explain what the sign of the result means.

$$\left[\frac{x^2}{2} - 2x \right]_0^2 = (2 - 4) - 0 = -2. \text{The negative sign means the curve lies below the } x\text{-axis on } [0, 2].$$
- 26 Find the (positive) area between $y = x - 2$ and the x -axis, from $x = 0$ to $x = 2$.
 The signed integral is -2 (from the previous question), so the area is $|-2| = 2$.
-

Properties of definite integrals

- 27 Given $\int_1^4 f(x) dx = 10$, find $\int_1^4 3f(x) dx$.
 By linearity, $\int_1^4 3f(x) dx = 3(10) = 30$.
- 28 Given $\int_0^2 f(x) dx = 5$ and $\int_2^5 f(x) dx = 7$, find $\int_0^5 f(x) dx$.
 By additivity over intervals: $5 + 7 = 12$.
- 29 Given $\int_0^3 f(x) dx = 8$, find $\int_3^0 f(x) dx$.
 Reversing the limits negates the integral: $\int_3^0 f(x) dx = -8$.
-

Finding a function from its derivative

- 30 A function satisfies $f'(x) = 6x^2 - 4$ and $f(1) = 5$. Find $f(x)$.

$$f(x) = 2x^3 - 4x + C. f(1) = 2 - 4 + C = 5 \Rightarrow C = 7. f(x) = 2x^3 - 4x + 7.$$
- 31 A function satisfies $f'(x) = 3x^2 + 2x$ and $f(0) = 1$. Find $f(x)$.

$$f(x) = x^3 + x^2 + C. f(0) = C = 1. f(x) = x^3 + x^2 + 1.$$
- 32 A function satisfies $f'(x) = 4x - 1$ and $f(2) = 10$. Find $f(x)$.

$$f(x) = 2x^2 - x + C. f(2) = 8 - 2 + C = 10 \Rightarrow C = 4. f(x) = 2x^2 - x + 4.$$
-

Average value of a function

- 33 Find the average value of $f(x) = x^2$ on $[0, 3]$, using $\text{avg} = \frac{1}{b-a} \int_a^b f(x) dx$.
 $\text{avg} = \frac{1}{3} \int_0^3 x^2 dx = \frac{1}{3}(9) = 3$.
- 34 Find the average value of $f(x) = 2x + 1$ on $[1, 5]$.
 $\int_1^5 (2x + 1) dx = [x^2 + x]_1^5 = 30 - 2 = 28$. $\text{avg} = \frac{28}{5-1} = 7$.

Word problems: integration in context

- 35 This question has two parts. A particle moves along a line with velocity $v(t) = 6t - 3t^2$ (m/s) for $0 \leq t \leq 4$.
- a Find the particle's displacement from $t = 0$ to $t = 2$, using $\int_0^2 v(t) dt$.
 $\int_0^2 (6t - 3t^2) dt = [3t^2 - t^3]_0^2 = 12 - 8 = 4$ meters.
- b Find the particle's displacement from $t = 0$ to $t = 4$, and explain why it is not simply twice the part (a) answer. $\int_0^4 (6t - 3t^2) dt = [3t^2 - t^3]_0^4 = 48 - 64 = -16$ meters. The velocity becomes negative for $t > 2$ (the particle reverses direction), so the displacements over $[0, 2]$ and $[2, 4]$ partially cancel rather than simply add.
- 36 This question has two parts. Water flows into a tank at a rate of $r(t) = 20 - 2t$ liters per minute, for $0 \leq t \leq 10$.
- a Write a definite integral for the total volume of water that flows in during the first 5 minutes.
 $V = \int_0^5 (20 - 2t) dt$.
- b Evaluate the integral to find the volume. $V = [20t - t^2]_0^5 = 100 - 25 = 75$ liters.

MCQ — Basic antiderivatives (the power rule)

- 37 Find $\int x^6 dx$.
- a. $\frac{x^6}{7} + C$
- b. $7x^5 + C$
- c. $\frac{x^7}{6} + C$
- d. $\frac{x^7}{7} + C$ ✓
- Power rule: $\frac{x^{6+1}}{6+1} + C$.

38 Find $\int 5x^3 dx$.

a. $5x^4 + C$

b. $\frac{5x^4}{4} + C$ ✓

c. $\frac{5x^4}{3} + C$

d. $15x^2 + C$

$5 \cdot \frac{x^4}{4} + C.$

39 Find $\int \sqrt[3]{x} dx$.

a. $\frac{3}{2}x^{2/3} + C$

b. $\frac{4}{3}x^{4/3} + C$

c. $\frac{3}{4}x^{4/3} + C$ ✓

d. $\frac{1}{3}x^{-2/3} + C$

$\int x^{1/3} dx = \frac{x^{4/3}}{4/3} + C = \frac{3}{4}x^{4/3} + C.$

40 Find $\int \frac{1}{x^4} dx$.

a. $\frac{1}{3x^3} + C$

b. $-\frac{1}{4x^3} + C$

c. $-\frac{1}{3x^3} + C$ ✓

d. $-\frac{4}{x^5} + C$

$\int x^{-4} dx = \frac{x^{-3}}{-3} + C = -\frac{1}{3x^3} + C.$

41 Find $\int 7 dx$.

a. C

b. $7x + C$ ✓

c. $7 + C$

d. $\frac{7x^2}{2} + C$

The antiderivative of a constant k is $kx + C$.

42 Find $\int \frac{1}{\sqrt{x}} dx$.

a. $2\sqrt{x} + C$ ✓

b. $\frac{1}{2}\sqrt{x} + C$

c. $-2\sqrt{x} + C$

d. $\sqrt{x} + C$

$$\int x^{-1/2} dx = \frac{x^{1/2}}{1/2} + C = 2\sqrt{x} + C.$$

MCQ — Antiderivatives of polynomials

43 Find $\int (6x^2 - 8x + 3) dx$.

a. $2x^3 - 4x^2 + 3x + C$ ✓

b. $2x^3 - 4x^2 + C$

c. $2x^3 - 8x^2 + 3x + C$

d. $12x - 8 + C$

Integrate each term.

44 Find $\int (5x + 9) dx$.

a. $\frac{5x^2}{2} + C$

b. $5x^2 + 9x + C$

c. $\frac{5x^2}{2} + 9x + C$ ✓

d. $\frac{5x^2}{2} + 9 + C$

Integrate each term.

45 Find $\int (x^2 - 6x + 5) dx$.

a. $\frac{x^3}{3} - 3x^2 + 5x + C$ ✓

b. $\frac{x^3}{3} - 6x^2 + 5x + C$

c. $\frac{x^3}{3} - 3x^2 + C$

d. $x^3 - 3x^2 + 5x + C$

Integrate each term.

46 Find $\int (3x^3 - 2x) dx$.

a. $3x^4 - x^2 + C$

b. $\frac{3x^4}{4} - x^2 + C$ ✓

c. $\frac{3x^4}{4} - 2x^2 + C$

d. $\frac{x^4}{4} - x^2 + C$

Integrate each term.

MCQ — Antiderivatives of trigonometric and exponential functions

47 Find $\int 2\cos x dx$.

a. $2\sin x + C$ ✓

b. $-2\sin x + C$

c. $2\cos x + C$

d. $\sin(2x) + C$

$\int \cos x dx = \sin x.$

48 Find $\int 3\sin x \, dx$.

- a. $3\sin x + C$
- b. $-3\sin x + C$
- c. $3\cos x + C$
- d. $-3\cos x + C$ ✓

$$\int \sin x \, dx = -\cos x.$$

49 Find $\int (e^x + 2) \, dx$.

- a. $e^x + 2 + C$
- b. $e^x + 2x + C$ ✓
- c. $xe^x + 2x + C$
- d. $e^x + C$

$$\int e^x \, dx = e^x; \int 2 \, dx = 2x.$$

50 Find $\int (\sin x + \cos x) \, dx$.

- a. $-\cos x - \sin x + C$
- b. $\cos x + \sin x + C$
- c. $\cos x - \sin x + C$
- d. $-\cos x + \sin x + C$ ✓

$$\int \sin x \, dx = -\cos x; \int \cos x \, dx = \sin x.$$

MCQ — Evaluating definite integrals

51 Evaluate $\int_0^3 2x \, dx$.

a. 9 ✓

b. 6

c. 3

d. 18

$$[x^2]_0^3 = 9.$$

52 Evaluate $\int_1^4 3x^2 \, dx$.

a. 21

b. 48

c. 64

d. 63 ✓

$$[x^3]_1^4 = 64 - 1 = 63.$$

53 Evaluate $\int_0^5 x \, dx$.

a. 5

b. 12.5 ✓

c. 25

d. 10

$$\left[\frac{x^2}{2}\right]_0^5 = 12.5.$$

54 Evaluate $\int_{-2}^1 4x^3 \, dx$.

a. -17

b. -15 ✓

c. 15

d. 1

$$[x^4]_{-2}^1 = 1 - 16 = -15.$$

55 Evaluate $\int_2^5 3 \, dx$.

a. 3

b. 9 ✓

c. 6

d. 15

$$3 \times (5 - 2) = 9.$$

56 Evaluate $\int_0^2 6x^2 \, dx$.

a. 16 ✓

b. 12

c. 24

d. 8

$$[2x^3]_0^2 = 16.$$

MCQ — Definite integrals with polynomial integrands

57 Evaluate $\int_0^3 (x^2 + 2) \, dx$.

a. 9

b. 15 ✓

c. 11

d. 21

$$\left[\frac{x^3}{3} + 2x \right]_0^3 = 9 + 6 = 15.$$

58 Evaluate $\int_1^4 (3x^2 - 2x + 1) dx$.

- a. 50
- b. 52
- c. 63
- d. 51 ✓

$$[x^3 - x^2 + x]_1^4 = 52 - 1 = 51.$$

59 Evaluate $\int_0^2 (4x^2 - 2x + 3) dx$.

- a. $\frac{32}{3}$
- b. $\frac{40}{3}$
- c. $\frac{38}{3}$ ✓
- d. 14

$$\left[\frac{4x^3}{3} - x^2 + 3x \right]_0^2 = \frac{32}{3} - 4 + 6 = \frac{38}{3}.$$

60 Evaluate $\int_{-1}^2 (x^2 - x) dx$.

- a. 4.5
- b. 2.5
- c. 0.5
- d. 1.5 ✓

$$\left[\frac{x^3}{3} - \frac{x^2}{2} \right]_{-1}^2 = \frac{2}{3} - \left(-\frac{5}{6} \right) = 1.5.$$

MCQ — Definite integrals of trigonometric and exponential functions

61 Evaluate $\int_0^{\pi/2} \sin x \, dx$.

a. $\frac{\pi}{2}$

b. -1

c. 0

d. 1 ✓

$$[-\cos x]_0^{\pi/2} = -0 + 1 = 1.$$

62 Evaluate $\int_0^{\pi} \cos x \, dx$.

a. π

b. -2

c. 0 ✓

d. 2

$$[\sin x]_0^{\pi} = 0 - 0 = 0.$$

63 Evaluate $\int_0^2 e^x \, dx$.

a. $e^2 - 1$ ✓

b. $2e$

c. $e^2 + 1$

d. e^2

$$[e^x]_0^2 = e^2 - 1.$$

MCQ — Area under a curve

64 Find the area under $y = x^2$ from $x = 0$ to $x = 4$, using a definite integral.

a. $\frac{64}{3}$ ✓

b. 64

c. 16

d. $\frac{16}{3}$
 $\left[\frac{x^3}{3}\right]_0^4 = \frac{64}{3}.$

65 Find the area under $y = 9 - x^2$ from $x = -3$ to $x = 3$.

a. 27

b. 54

c. 36 ✓

d. 18

$\left[9x - \frac{x^3}{3}\right]_{-3}^3 = 18 - (-18) = 36.$

66 The graph shows $y = x^2$. Find the area under the curve from $x = 0$ to $x = 3$, using a definite integral.

a. 9 ✓

b. 18

c. 3

d. 27

$\left[\frac{x^3}{3}\right]_0^3 = 9.$

67 The graph shows $y = 4 - x^2$. Find the area under the curve from $x = -2$ to $x = 2$.

a. $\frac{16}{3}$

b. 32

c. 16

d. $\frac{32}{3}$ ✓
 $\left[4x - \frac{x^3}{3}\right]_{-2}^2 = \frac{16}{3} - \left(-\frac{16}{3}\right) = \frac{32}{3}.$

MCQ — Signed area and regions below the axis

68 Evaluate $\int_0^3 (x - 3) dx$ and explain what the sign of the result means.

- a. 4.5; the region lies above the x -axis
- b. -4.5; the region lies above the x -axis
- c. 4.5; the region lies below the x -axis
- d. -4.5; the region lies below the x -axis ✓

$$\left[\frac{x^2}{2} - 3x \right]_0^3 = 4.5 - 9 = -4.5; \text{ negative means the region lies below the } x\text{-axis.}$$

69 Find the (positive) area between $y = x - 3$ and the x -axis, from $x = 0$ to $x = 3$.

- a. 9
- b. -4.5
- c. 3
- d. 4.5 ✓

Take the absolute value of the signed integral: $|-4.5| = 4.5$.

70 Evaluate $\int_0^4 (x - 4) dx$ and state whether the region is above or below the x -axis.

- a. 8; above
- b. -8; above
- c. 8; below
- d. -8; below ✓

$$\left[\frac{x^2}{2} - 4x \right]_0^4 = 8 - 16 = -8; \text{ negative means below the } x\text{-axis.}$$

MCQ — Properties of definite integrals

71 Given $\int_2^6 f(x) dx = 12$, find $\int_2^6 4f(x) dx$.

a. 12

b. 16

c. 48 ✓

d. 3

$4 \times 12 = 48$.

72 Given $\int_0^3 f(x) dx = 6$ and $\int_3^7 f(x) dx = 9$, find $\int_0^7 f(x) dx$.

a. 13

b. 3

c. 15 ✓

d. 54

$6 + 9 = 15$.

73 Given $\int_0^5 f(x) dx = 14$, find $\int_5^0 f(x) dx$.

a. 14

b. $\frac{1}{14}$

c. 0

d. -14 ✓

Reversing the limits negates the integral: -14 .

74 Given $\int_1^5 f(x) dx = 20$ and $\int_1^5 3 dx = 12$, find $\int_1^5 [f(x) - 3] dx$.

a. -8

b. 8 ✓

c. 32

d. 17

$20 - 12 = 8$.

75 Given $\int_0^4 f(x) dx = 9$ and $\int_0^4 g(x) dx = 5$, find $\int_0^4 [f(x) + g(x)] dx$.

a. $\frac{9}{5}$

b. 45

c. 4

d. 14 ✓

$9 + 5 = 14$.

MCQ — Finding a function from its derivative

76 A function satisfies $f'(x) = 8x^3 - 6$ and $f(1) = 3$. Find $f(x)$.

a. $2x^4 - 6x + 7$ ✓

b. $2x^4 - 6x + 3$

c. $8x^4 - 6x + 7$

d. $2x^4 - 6x - 7$

$f(x) = 2x^4 - 6x + C$; $f(1) = 2 - 6 + C = 3 \Rightarrow C = 7$.

77 A function satisfies $f'(x) = 4x^2 + 3x$ and $f(0) = 2$. Find $f(x)$.

a. $\frac{4x^3}{3} + 3x^2 + 2$

b. $\frac{4x^3}{3} + 1.5x^2$

c. $\frac{4x^3}{3} + 1.5x^2 + 2$ ✓

d. $4x^3 + 1.5x^2 + 2$

$f(x) = \frac{4x^3}{3} + 1.5x^2 + C$; $f(0) = C = 2$.

78 A function satisfies $f'(x) = 6x - 5$ and $f(3) = 20$. Find $f(x)$.

a. $3x^2 - 5x + 20$

b. $3x^2 - 5x + 8$ ✓

c. $6x^2 - 5x + 8$

d. $3x^2 - 5x - 8$

$$f(x) = 3x^2 - 5x + C; f(3) = 27 - 15 + C = 20 \Rightarrow C = 8.$$

79 A function satisfies $f'(x) = 2x + 7$ and $f(-1) = 4$. Find $f(x)$.

a. $x^2 + 7x + 10$ ✓

b. $2x^2 + 7x + 10$

c. $x^2 + 7x - 10$

d. $x^2 + 7x + 4$

$$f(x) = x^2 + 7x + C; f(-1) = 1 - 7 + C = 4 \Rightarrow C = 10.$$

MCQ — Average value of a function

80 Find the average value of $f(x) = x^2$ on $[0, 4]$, using $\text{avg} = \frac{1}{b-a} \int_a^b f(x) dx$.

a. $\frac{64}{3}$

b. 16

c. $\frac{16}{3}$ ✓

d. 4

$$\frac{1}{4} \left[\frac{x^3}{3} \right]_0^4 = \frac{1}{4} \times \frac{64}{3} = \frac{16}{3}.$$

81 Find the average value of $f(x) = 3x + 2$ on $[0, 4]$.

a. 32

b. 4

c. 14

d. 8 ✓

$$\frac{1}{4}[1.5x^2 + 2x]_0^4 = \frac{1}{4}(32) = 8.$$

82 Find the average value of $f(x) = x^2 - 1$ on $[1, 3]$.

a. $\frac{8}{3}$

b. $\frac{10}{3}$ ✓

c. $\frac{20}{3}$

d. 6

$$\frac{1}{2}\left[\frac{x^3}{3} - x\right]_1^3 = \frac{1}{2}\left(6 - \left(-\frac{2}{3}\right)\right) = \frac{10}{3}.$$

MCQ — Word problems: integration in context

83 A particle moves along a line with velocity $v(t) = 8t - 2t^2$ (m/s) for $0 \leq t \leq 4$. Find the particle's displacement over this interval.

a. 64

b. $\frac{64}{3}$ ✓

c. 32

d. $\frac{192}{3}$

$$\int_0^4 (8t - 2t^2) dt = \left[4t^2 - \frac{2t^3}{3}\right]_0^4 = \frac{64}{3} \text{ m.}$$

- 84 Water flows into a tank at a rate of $r(t) = 30 - 3t$ liters per minute, for $0 \leq t \leq 10$. Find the total volume of water that flows in during this time.

a. 300

b. 150 ✓

c. 450

d. 100

$$\int_0^{10} (30 - 3t) dt = [30t - 1.5t^2]_0^{10} = 150 \text{ liters.}$$

- 85 The graph shows a particle's velocity $v(t) = 8t - 2t^2$ for $0 \leq t \leq 4$. Find the particle's displacement (the area under the curve).

a. $\frac{64}{3}$ ✓

b. 16

c. 32

d. 64

$$\int_0^4 (8t - 2t^2) dt = \frac{64}{3} \text{ m.}$$

- 86 A factory's production rate is $p(t) = 50 + 4t$ items per hour, for $0 \leq t \leq 8$. Find the total number of items produced during this 8-hour period.

a. 528 ✓

b. 656

c. 450

d. 400

$$\int_0^8 (50 + 4t) dt = [50t + 2t^2]_0^8 = 528 \text{ items.}$$